

PROJECT STATUS / 09 SEPTEMBER 2026

MedVault

What is still incomplete

A practical completion checklist for Phase 1 and features 15-21, compared with the newly supplied scope. Future features 22-28 are listed separately.

Bottom line

The foundation and Phase 2 have substantial implementation and test evidence. **Phase 2 is not live-accepted, and Phase 3 is not an operational medication product.** Existing models and API methods must not be counted as completed user workflows.

SCOPE	CURRENT POSITION	MAIN REMAINING WORK
Phase 1	Implemented foundation; live acceptance incomplete	Recheck screening, hosted persistence, private files, recovery and cross-account access.
15-16 / Phase 2	Comparisons, episodes and charts implemented; some expanded UI requirements partial	Hosted/browser acceptance, per-metric latest/history experience and context limitations.
17-18 / Phase 2	Summary code implemented; live blocked. Urgency automation disabled.	Safe hosted migration, working Gemini generation, live summaries and supported guidance boundary.
19 / Phase 3	Medication models and basic API only	Usable medication UI, validation, history, ownership fixes and persistence tests.
20-21 / Phase 3	Scheduling models/basic API exist; reminder and safety workflows missing	Delivery engine/calendar, duplicate-safe logging, typed safety adapter, source selection and tests.

Three immediate blockers

1. Hosted schema/history reconciliation and a verified backup are still required.
2. Gemini generation returned HTTP 404 in the last live check.
3. Real hosted browser/cross-device acceptance is not recorded.

Status key: Missing = no usable implementation found. Partial = scaffolding or only part of the requirement exists. Blocked = an external/safety gate prevents completion. Unverified = implementation exists but required evidence is absent.

Evidence date: 2026-09-09. This is a read-only source/evidence review plus PDF creation, not a new implementation or live-service certification. Sources [S1]-[S6] are identified on page 7.

PHASE 1 / MILESTONE A

Close the live-evidence gaps

Do not rebuild the existing foundation. The remaining work is mainly controlled live verification and repair of any defects those checks expose. A missing live pass does not mean the code is missing. [S2, S3, S5]

ID / STATUS	WHAT REMAINS AND HOW TO CLOSE IT
A-01 UNVERIFIED	Same-account hosted persistence Use two fresh authenticated browser sessions, then the actual Mac and Windows devices. Verify the same profile, documents, reviewed values and patient mapping; confirm patient creation remains idempotent.
A-02 UNVERIFIED	Complete upload and screening matrix Run synthetic readable-report, prescription, unrelated, unreadable/uncertain and unsupported-file cases through the real API/worker/provider. Prove screening precedes extraction and a checker outage produces a retryable technical failure.
A-03 UNVERIFIED	Private Storage and ownership Verify metadata-to-object consistency, authorized signed preview/download and denial for a second account. Include report reads, verification, retry, deletion and linked IDs; do not expose actual patient data in evidence.
A-04 UNVERIFIED	Failure and recovery behavior Exercise worker shutdown/restart, enqueue failure, duplicate jobs, timeouts and reconciliation. Compare queue, database, API and browser states; prove there is no false success or stale overwrite.
A-05 BLOCKED	Backup, migration history and grants Create a protected hosted backup and prove restoration in isolation before history reconciliation. Review actual grants/RLS and the server database role. Existing successful API tests do not prove Data API isolation.
A-06 UNVERIFIED	Operational acceptance and privacy gates Recheck running-process environment alignment, health/readiness behavior, safe errors/logs, Auth lifecycle, retention/audit controls and provider privacy approvals. Treat documents describing a plan as plans, not completed operations.

Already evidenced - do not redo unnecessarily

The last checkpoint records matching API/worker configuration and a Redis PONG. Phase 1 regression tests passed. Screening code exists, so describing protection as only a client-side label would be stale. Live screening and end-to-end extraction still need evidence.

Close this milestone when: the real hosted workflow, failure matrix and cross-account checks have dated evidence, and every actual defect discovered is fixed and retested. [S2, S3]

PHASE 2 / FEATURES 15-18

Finish hosted operation and acceptance

Comparison, episode, chart and summary code should be reused. The connected rehearsal used real Redis and an isolated PostgreSQL database, but **mocked Gemini and controlled authentication**. It was not a hosted or browser-session pass. [S2, S3]

ID / STATUS	WHAT REMAINS AND HOW TO CLOSE IT
B-01 BLOCKED	Hosted summary persistence The last metadata check found only the initial Prisma migration recorded and no medical_summaries table. Episode/medication tables already exist outside that history. Reconcile only after backup/restore verification, then apply the reviewed additive migration and validate counts, relationships and grants.
B-02 BLOCKED	Gemini generation Configured gemini-2.5-flash returned HTTP 404 through SDK and direct REST generation, although model listing succeeded. Resolve the model/project/access inconsistency privately and prove one real single-report and one episode summary. Do not guess that the key is invalid or silently change models.
B-03 UNVERIFIED	Real hosted summary lifecycle Verify request, enqueue, worker pickup, real provider response, grounded persistence, API reload and browser display. Repeat after corrections/membership changes; prove old worker results cannot become current and cached summaries avoid duplicate calls.
B-04 UNVERIFIED	Interactive charts and recovery Exercise touch/keyboard access, source links, missing ranges/dates, empty/one-point states, retry, delayed polling and fresh-account cache isolation in a browser. Repeat a real worker restart; current tests are not interactive browser evidence.
B-05 PARTIAL	Expanded history/cards and explanation scope Dedicated per-metric latest-value cards/test-history workflow are not evidenced by the current episode chart/table UI. Summaries select deterministic fact text; a sourced plain-English terminology/explanation layer requested by the new scope remains to be completed or explicitly agreed.
B-06 PARTIAL / BLOCKED	Clinical context and attention rules Laboratory/hospital context exists, but method/specimen context is not persisted for comparison. Keep uncertainty explicit. The cannot-assess state exists; a reusable validated-rule interface, sourced rule fixtures and clinical approval are not evidenced. Automated urgency must remain disabled.

Do not list as missing: report summary routes, episode editing/membership, basic trend/table components, grounded fact validation, source fingerprints and stale-write guards already exist. Their hosted verification is what remains. [S2, S3, S5]

PHASE 3 / FEATURE 19 / MILESTONE D

Medication records are only partial

Medication, MedicationSchedule, ScheduleOccurrence, IntakeLog and ReminderDelivery models exist, along with basic medication/schedule/intake API methods. No medication-facing web routes were found in the inspected route inventory. [S4, S5]

ID / STATUS	WHAT REMAINS AND HOW TO CLOSE IT
D-01 MISSING	Complete patient-facing medication UI Build create/view/edit/archive flows, prescription selection, current/past statuses, useful empty/errors and accessible forms. Add medication list pagination/filtering; the current list API returns an unpaginated array.
D-02 PARTIAL	Strength, dose, quantity and regimen semantics Strength, dose amount and dose unit are separate fields, but distinct administration-quantity handling is not evidenced. Support patient-entered instructions without inventing a regimen. Define three-times-daily versus every-eight-hours, as-needed behavior and after-meal instructions; current meal enum lacks AFTER_MEAL.
D-03 CONFIRMED GAP	Prescription ownership on medication edits Creation checks the linked prescription owner/type. updateMedication accepts the replacement prescriptionId without repeating that check. Apply the same validation during edits and add a regression test proving a foreign prescription cannot be linked.
D-04 PARTIAL	History and schedule lifecycle Current medication updates overwrite fields; immutable edit/status history is not evidenced. Archive changes medication status only. Implement versioned changes and cancel/supersede future work on pause/stop/archive while retaining earlier dose meaning and records.
D-05 PARTIAL	Time, date and instruction validation Current HH:mm validation is shape-only, the timezone is only a nonempty string, and date ordering/interval-specific data are incomplete. Reject invalid clock times, unknown IANA zones, reversed dates and unsupported combinations. Validate before persistence and show useful form errors.
D-06 PARTIAL / UNVERIFIED	Resolution and acceptance Keep entered names unresolved unless a supported source establishes an ingredient; no resolver integration is evidenced. Add medication/schedule ownership, history and fresh-session tests, then prove hosted persistence and prescription links with real authentication.

Source-review finding, not a new implementation

The ownership and validation gaps above were observed in the current code while preparing this report. They have not been repaired or exploit-tested in this PDF-only task. Do not enable medication workflows as complete on the basis of their database models.

PHASE 3 / FEATURE 20 / MILESTONE E

Reminder execution and tracking remain

Basic schedule creation/deactivation and intake endpoints are scaffolding. A queue-name constant and delivery table do not constitute a running reminder system. The worker currently registers report extraction and summary processing, not medication delivery. [S4, S5]

ID / STATUS	WHAT REMAINS AND HOW TO CLOSE IT
E-01 MISSING	Durable occurrence generation Implement a bounded rolling horizon from PostgreSQL schedules, confirmed IANA timezone handling, UTC instants/local intent, midnight/end-date rules, daylight-saving behavior and explicit as-needed exclusions.
E-02 MISSING	Queue producer, consumer and reconciliation Create reminder jobs and a worker with duplicate-safe claims, schedule-version checks, bounded retries and persisted attempt/delivery results. On startup, rebuild missing due work without replaying obsolete reminders or flooding users with historical notifications.
E-03 PARTIAL	Duplicate-safe intake and correction history The current logIntake method creates a new log for each request, then updates occurrence status. Add request idempotency, revision/cancellation checks and correction history so retries or two devices cannot create conflicting dose records.
E-04 MISSING	Calendar and dose-history UI Provide daily/weekly schedules, upcoming and overdue/unacknowledged entries, taken/skipped actions, correction flows and history. An unacknowledged reminder must not be presented as proof of a missed dose.
E-05 MISSING	In-app and optional browser notifications Persist in-app reminders; request browser permission only on intentional action. Handle denial and unsupported browsers, keep medical details out of previews by default, and never mark an unsuccessful delivery as delivered.
E-06 MISSING	Refill and course-ending reminders Use explicit stock, dose and actual taken events for qualified estimates; do not subtract skipped doses or imply accuracy with missing data. Course-ending messages must not recommend extending or altering treatment.
E-07 UNVERIFIED	Scheduling acceptance suite Add deterministic-clock, timezone/DST, two-worker, restart, stale-job cancellation, duplicate intake, permission-denial and refill tests. Verify same-account cross-session synchronization and different-account denial on the hosted setup.

Runtime limitation: the API and worker remain local. No running worker means delivery cannot be guaranteed; browser/system notifications also depend on support and permission. Document this honestly instead of deploying services without a separate request.

PHASE 3 / FEATURE 21 / MILESTONE F

Medication safety is not integrated

No medication-safety adapter, result workflow or UI was found in the inspected Phase 3 paths. The missing provider is not a reason to skip the typed boundary and honest unavailable state; those can be implemented independently. [S1, S4, S5]

REQUESTED CHECK	WHAT STILL HAS TO EXIST
Duplicate medicine / ingredient	Confident ingredient resolution, active-period/formulation comparison and explicit unresolved-name/partial-coverage outcomes.
Drug-drug interactions	An available, licensed, authoritative source with documented coverage, severity provenance and update/version information; not Gemini or terminology lookup alone.
Allergy conflicts	User-reported allergy context plus supported ingredient/class mapping. No guessed cross-reactivity.
Condition conflicts	A sourced contraindication/precaution rule and matching patient context; missing context must stay unknown.
Dosage conflicts	Separate inconsistent entered schedules from clinical dose judgments. Clinical checks need validated rules and every required input; no fabricated thresholds.

Implementation still required

Adapter and contracts: typed requests/results, minimal transmitted data, bounded timeout/retry/cache behavior, safe errors and recorded source/version/date.

Persistence and invalidation: store sourced findings, coverage and input revisions; invalidate after medicine, allergy or condition changes and reject stale responses.

Usable states: findings detected; no listed finding within checked coverage; partial coverage; unknown medicine; missing context; service unavailable. Never collapse these into an unqualified "safe".

Evidence: sourced fixture tests for each supported check, malformed/provider-failure tests, ownership tests, then a real tested source before any live-integration claim.

Provider decision needed before integration

Prepare a concrete source option with licensing, cost, geographic/brand coverage, update policy, intended-use limits and required access. Obtain approval before any new paid service. Until then, keep affected clinical checks blocked and visible as unavailable - not powered by hidden mock results.

Gemini may explain a supported sourced finding, but must not invent interaction facts, doses, contraindications or clinical severity. No automatic medication changes or dosing advice belong in this workflow. This report makes no provider-selection or clinical-validation claim.

EXECUTION ORDER / EVIDENCE

What to do next

1. **Protect and reconcile.** Approve and verify a hosted backup/restore checkpoint; reconcile existing schema history before applying additive changes. Review grants and the runtime role.
2. **Restore live AI and close acceptance.** Resolve Gemini generation, then complete hosted Phase 1/2 and browser/cross-device evidence.
3. **Finish medication records and schedules.** Address edit ownership, validation and history first; then build the UI, occurrence engine, delivery, logging and calendar.
4. **Add the safety boundary.** Implement typed unavailable/partial states and fixtures while the provider/clinical approval decision is pending; integrate only supported checks.

Future features 22-28 - deferred, not Phase 1-3 blockers

GROUP	CURRENT ASSESSMENT / DEFERRED WORK
22 / Family	Partial: emergency-contact profile fields exist; dependent/guardian workflows and a dedicated emergency profile are not evidenced.
23 / Sharing	Missing workflow: selected-report links, expiry, revocation, permissions and access history.
24 / Doctor portal	Missing workflow: consent, doctor review/notes and communication.
25 / Broader alerts	Partial status display; broad notification/email/push channels not evidenced. Medication reminders remain required now, not deferred.
26 / Exports/recovery	Partial: document deletion and recovery plans exist; data export/account-wide controls and verified backup/restore remain incomplete. Core backup gates still apply now.
27 / Organization	Partial: episodes and filters exist; tags/folders/favorites and advanced grouping remain deferred.
28 / Administration	Partial: health/logging diagnostics exist; admin/support, security analytics and cost dashboards remain deferred.

Evidence and review limits

Last recorded suite: **263 tests passed**; one opt-in SQL rehearsal passed separately. Lint, typecheck, Prisma validation and the local-opt-in build passed. These checks were not rerun for this PDF-only request. The connected rehearsal used real Redis and synthetic PostgreSQL with mock Gemini/controlled Auth; no hosted or actual Mac-to-Windows pass is claimed.

[S1] User attachment: "MedVault: Verify Phase 1 and complete Phases 2 and 3", supplied in this task.

[S2] docs/PHASE_PROGRESS.md, current checkpoint 2026-09-09.

[S3] docs/IMPLEMENTATION_REVIEW.md, verification ledger and operational gates.

[S4] packages/database/prisma/schema.prisma; packages/shared/src/index.ts.

[S5] apps/api/src/services/prisma-service.ts; apps/api/src/app.ts; apps/worker/src/index.ts; current web routes/components and AI adapter.

[S6] docs/BACKUP_AND_RECOVERY.md; existing privacy/security plans. Plans are not proof of operational approval.